

Sauti

Multilingual AI Voice Agent Platform

Software Requirements Specification

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Table of Contents

1.	Introduction & Project Overview	3
2.	Product Vision & Goals	4
3.	System Architecture Overview	5
4.	Functional Requirements	7
4.1	Tenant Management	7
4.2	Agent Configuration	8
4.3	Real-Time Call Pipeline	9
4.4	Language & Voice Engine	10
4.5	Calendar & Booking Integration	11
4.6	Analytics & Reporting	12
4.7	Billing & Subscription	13
5.	Non-Functional Requirements	14
6.	Technology Stack	15
7.	Data Models & Database Schema	16
8.	API Specification	18
9.	Security Requirements	20
10.	Infrastructure & Deployment	21
11.	Build Sequence & Milestones	22
12.	Risks & Mitigations	23
13.	Glossary	24

1. Introduction & Project Overview

This document is the Software Requirements Specification (SRS) for **Sauti** — a white-label, multilingual AI voice agent platform targeting businesses in French-speaking Africa, the MENA region, and East Africa (Swahili-speaking markets). The name Sauti means *voice* in Swahili.

Sauti enables any business — regardless of industry — to deploy a customized AI phone agent that answers calls, converses naturally in Arabic, French, Swahili, or English, and completes tasks such as booking appointments, answering FAQs, and routing inquiries — 24 hours a day, 7 days a week.

1.1 Purpose of This Document

This SRS defines the complete functional and non-functional requirements for the Sauti platform. It is intended to guide the solo developer (Zacharie) through the full build sequence, from MVP to production-ready release. It covers the Spring Boot backend, Flutter Web dashboard, real-time call pipeline, and third-party integrations.

1.2 Scope

Sauti consists of three major components:

- **Spring Boot Backend (Modular Monolith)** — Core platform serving all business logic: tenant management, agent configuration, call orchestration, analytics, and billing.
- **Flutter Web Dashboard** — The self-service portal where businesses create and configure their voice agents, view call analytics, and manage billing.
- **Real-Time Call Pipeline** — The latency-sensitive subsystem handling live phone calls via Twilio Media Streams, chaining STT → Language Detection → LLM → TTS in under 400ms.

1.3 Definitions

Term	Definition
Tenant	A registered business using the Sauti platform.
Agent	An AI voice agent configured by a tenant for a specific phone number.
Turn	One exchange in a phone conversation: caller speaks → agent responds.
STT	Speech-to-Text: converts audio to transcript text.
TTS	Text-to-Speech: converts LLM response text to audio.
LLM	Large Language Model: the AI model generating conversational responses.
RAG	Retrieval-Augmented Generation: grounding LLM answers in tenant-provided knowledge.
TTFT	Time-to-First-Token: latency from end of caller speech to first audio byte returned.

2. Product Vision & Goals

"Any business. Any language. Any call." — Sauti gives every business in Africa and the MENA region a tireless, intelligent phone agent that speaks their customers' language.

2.1 Problem Statement

Businesses in French-speaking Africa, the Arab world, and East Africa lose revenue daily from missed calls, slow response times, and the inability to afford human receptionists around the clock. Existing AI voice agent platforms (Retell AI, Bland AI, VoiceDesk AI) are built exclusively for English-speaking Western markets. No quality solution exists natively for Arabic, French, or Swahili-speaking businesses.

2.2 Target Markets

Market	Languages	Example Customers	Key Use Case
Francophone Africa (Senegal, Côte d'Ivoire, Cameroon, Morocco)	French, Wolof, Darija	Salons, clinics, schools, real estate agencies	Appointment booking, FAQ answering
MENA (Egypt, Algeria, Saudi Arabia)	Arabic (MSA + dialects), French	Clinics, pharmacies, legal firms, banks	Call routing, patient follow-up, reminders
East Africa (Kenya, Tanzania, Uganda)	Swahili, English	Microfinance, FMCG distributors, schools	Payment reminders, order status, booking

2.3 Business Goals

- BG-01** Generate \$2,000 MRR within 12 months of launch
- BG-02** Onboard 20 paying tenants within 6 months
- BG-03** Support French, Arabic, and Swahili at launch
- BG-04** Achieve TTFT (Time-to-First-Token) under 400ms per turn
- BG-05** Operate at under \$100/month infrastructure cost at 50 tenants
- BG-06** Allow any industry to self-configure an agent without developer help

3. System Architecture Overview

Sauti is built as a **modular monolith** on Spring Boot. All modules are deployed as a single JAR, but are internally separated by strict package boundaries. Each module communicates with others only through defined service interfaces, not direct repository access. This enables future extraction into microservices without a full rewrite.

3.1 Modular Monolith Structure

Module	Responsibility	Key Dependencies
module-auth	JWT auth, OAuth (Google/Apple), user registration, password reset	Spring Security, Redis
module-tenant	Tenant (business) account management, plan enforcement, subscription	PostgreSQL
module-agent	Agent configuration storage: language, voice, system prompts, schedule	PostgreSQL, Redis
module-call	Call orchestration: Twilio webhook handling, WebSocket connection	Twilio SDK, WebSockets
module-nlp	Language detection, STT provider routing, TTS provider selection	Deepgram, Intron, ElevenLabs
module-llm	LLM prompt construction from agent config + conversation history	OpenAI, Gemini, StreamingResponse
module-calendar	Calendar integration: Google Calendar, Calendly, generic booking API	Google Calendar API
module-analytics	Call log aggregation, duration, sentiment, success rate per agent	PostgreSQL, Spring Scheduler
module-billing	Subscription plans, usage metering (minutes), Paddle/LemonSqueezy	Redis, PostgreSQL
module-api	REST controllers, request/response DTOs, OpenAPI documentation	Spring Web

3.2 Real-Time Call Pipeline

The call pipeline is the most latency-sensitive part of the system. Each turn must complete in under 400ms to feel natural. The pipeline operates as follows:

- Step 1

Caller dials the tenant's Twilio number → Twilio fires a webhook to Spring Boot → module-call opens a WebSocket connection to Twilio Media Streams to receive raw audio.
- Step 2

Audio chunks stream over WebSocket to module-nlp → forwarded to Deepgram (or Intron Sahara for Swahili) in real time → transcript returned with VAD (voice activity detection) end-of-speech signal.
- Step 3

Transcript sent to language detector → language identified from first turn or from agent default → correct LLM prompt template selected.
- Step 4

module-llm builds prompt: tenant system prompt + conversation history (Redis) + caller transcript → sends to Gemini 2.5 Flash / GPT-4o → streams token response.
- Step 5

Tokens forwarded to ElevenLabs TTS as they arrive (streaming TTS) → audio bytes returned and immediately forwarded back over Twilio Media Streams WebSocket to caller.
- Step 6

Turn logged: transcript, duration, language, sentiment → stored in PostgreSQL via module-analytics.

Latency budget per turn: STT $\leq 80\text{ms}$ · Language detect $\leq 10\text{ms}$ · LLM first token $\leq 200\text{ms}$ · TTS first chunk $\leq 80\text{ms}$ · Network $\leq 30\text{ms}$ = Total $\leq 400\text{ms}$ TTFT

4. Functional Requirements

Requirements are tagged by priority: **High** (MVP — must have at launch), **Medium** (v1.1 — within 3 months), **Low** (future roadmap). Each requirement is assigned to a backend module.

4.1 Tenant Management (module-tenant)

ID	Requirement	Priority	Module
TM-01	Businesses register with email, business name, country, and phone number	High	module-tenant
TM-02	Email verification required before agent activation	High	module-auth
TM-03	Tenant dashboard shows account status, active agents, and plan details	High	module-tenant
TM-04	Support multiple agents per tenant (e.g. one per branch)	Medium	module-tenant
TM-05	Tenant can invite team members with role-based access (Admin / Viewer)	Low	module-tenant
TM-06	Tenant can delete their account and all associated data (GDPR)	Medium	module-tenant

4.2 Agent Configuration (module-agent)

ID	Requirement	Priority	Module
AC-01	Each agent is linked to exactly one Twilio phone number provisioned on the platform	High	module-agent
AC-02	Tenant sets agent name, greeting message, and language(s) to support	High	module-agent
AC-03	Tenant selects a TTS voice from a library (language-appropriate voices per provider)	High	module-agent
AC-04	Tenant writes a system prompt describing agent behavior, tone, and knowledge	High	module-agent
AC-05	Tenant can upload an FAQ document (PDF/text); RAG pipeline indexes for grounding	Medium	module-agent
AC-06	Agent operating hours configurable (e.g. only active Mon–Fri 8am–6pm); plays voice message outside hours	Medium	module-agent
AC-07	Tenant can test agent via a web-based call simulator before going live	Medium	module-agent
AC-08	Agent configuration changes apply immediately without call interruption	High	module-agent
AC-09	Tenant can duplicate an agent configuration to create a new agent quickly	Low	module-agent
AC-10	Tenant defines escalation trigger phrases (e.g. 'speak to a human') to transfer the call	High	module-agent

4.3 Real-Time Call Pipeline (module-call)

ID	Requirement	Priority	Module
CP-01	Platform provisions a Twilio phone number per agent at agent creation	High	module-call
CP-02	Inbound call triggers webhook → WebSocket media stream opened within 200ms	High	module-call
CP-03	Audio streamed to STT provider in real time; transcript returned with HVD signal	High	module-call
CP-04	Conversation history stored in Redis per call session; cleared after call ends	High	module-call
CP-05	LLM response tokens forwarded to TTS in streaming fashion (no wait for full response)	High	module-call
CP-06	Call recording saved to Firebase Storage with tenant consent flag checked	Medium	module-call
CP-07	Human transfer: platform bridges call to tenant's human number via Twilio transfer	High	module-call
CP-08	Voicemail: if agent cannot handle query, offers to take a message; stores transcription	Medium	module-call
CP-09	Outbound calling: platform can initiate calls (reminders, follow-ups) via Twilio	Medium	module-call
CP-10	Max concurrent calls per tenant enforced by plan tier; excess calls get busy signal	High	module-billing

4.4 Language & Voice Engine (module-nlp)

ID	Requirement	Priority	Module
LV-01	Supported languages at launch: French, Arabic (MSA + Egyptian dialect), Swahili, English	High	module-nlp
LV-02	Language auto-detected from first caller utterance using STT confidence scores	High	module-nlp
LV-03	Agent responds in detected language regardless of tenant default if multilingual enabled	Medium	module-nlp
LV-04	Swahili STT routes to Intron Sahara v2 API; all other languages route to Deepgram	High	module-nlp
LV-05	TTS voice selection is language-aware: French → ElevenLabs FR voice, Arabic → Azure AR	High	module-nlp
LV-06	Code-switching mid-call (caller switches language) triggers voice/prompts re-selection	High	module-nlp
LV-07	LLM provider selection configurable: Gemini 2.5 Flash (default, best African language support), GPT-4o (fallback)	High	module-nlp
LV-08	Tenant can provide custom vocabulary list for domain-specific terms (e.g. medical names, products)	Medium	module-nlp

4.5 Calendar & Booking Integration (module-calendar)

ID	Requirement	Priority	Module
CB-01	Agent can check real-time availability from tenant's Google Calendar	High	module-calendar
CB-02	Agent can create a calendar event (booking) including caller name, phone, and service type	High	module-calendar
CB-03	Calendly webhook integration as alternative to Google Calendar	Medium	module-calendar
CB-04	Generic outbound webhook: POST booking details to tenant's custom endpoint	Medium	module-calendar
CB-05	Agent sends SMS confirmation to caller after successful booking via Twilio	Medium	module-call
CB-06	Agent handles rescheduling: can cancel existing booking and create new one	Medium	module-calendar
CB-07	Tenant sets booking buffer time (e.g. 15-min gap between appointments)	Low	module-calendar
CB-08	Agent reads back available slots to caller and confirms chosen slot verbally	High	module-call

4.6 Analytics & Reporting (module-analytics)

ID	Requirement	Priority	Module
AN-01	Call log: every call stored with timestamp, duration, language, outcome, transcript	High	module-analytics
AN-02	Dashboard metrics: total calls, answered vs missed, avg duration, calls by language	High	module-analytics
AN-03	Call outcome classification: Booking made / FAQ answered / Human transferred / Voice mail	High	module-analytics
AN-04	Tenant can search and replay any call transcript from the dashboard	Medium	module-analytics
AN-05	Weekly summary email auto-sent to tenant: calls handled, bookings made, peak hours	Medium	module-analytics
AN-06	Caller sentiment analysis per call (positive/neutral/negative) via LLM	Low	module-analytics
AN-07	Peak call hours heatmap (hour × day of week) displayed on dashboard	Medium	module-analytics

4.7 Billing & Subscription (module-billing)

Payment processor: Lemon Squeezy or Paddle — both support payments from Egypt and are compatible with your target markets without requiring a Stripe Atlas entity.

ID	Requirement	Priority	Module
BL-01	Three subscription plans: Starter (\$49/mo), Growth (\$99/mo), Pro (\$199/mo)	High	module-billing
BL-02	Each plan includes a monthly minute allowance; overage charged per minute	High	module-billing
BL-03	Usage metered per tenant in Redis; synced to PostgreSQL hourly	High	module-billing
BL-04	Tenant receives email warning at 80% and 100% of minute allowance	High	module-billing
BL-05	Agent auto-pauses when monthly limit reached; resumes on plan upgrade or next cycle	High	module-billing
BL-06	Billing webhook from Lemon Squeezy updates tenant plan in real time	High	module-billing

ID	Requirement	Priority	Module
BL-07	Tenant can view current usage and invoices from the dashboard	Medium	module-billing
BL-08	14-day free trial on all plans; no credit card required for trial	Medium	module-billing

5. Non-Functional Requirements

5.x Performance

- TTFT (Time-to-First-Token) per call turn: $\leq 400\text{ms}$ (P95)
- API response time for dashboard operations: $\leq 500\text{ms}$ (P95)
- WebSocket connection establishment: $\leq 200\text{ms}$
- Concurrent calls per server instance: ≥ 50 without degradation

5.x Scalability

- Single DigitalOcean droplet (\$12/mo) must handle ≥ 20 concurrent tenants at MVP
- Database must support ≥ 500 tenants without schema changes
- Architecture must allow extraction of module-call into a separate service when concurrent calls exceed 100

5.x Reliability

- Platform uptime target: 99.5% monthly (allows ~ 3.6 hours downtime/month)
- Failed call turns must retry STT/LLM once before falling back to 'I didn't catch that' response
- Twilio webhook must respond within 5 seconds or Twilio will retry — Spring Boot must ack immediately and process async

5.x Security

- All API endpoints require JWT authentication except `/auth/**` and Twilio webhook signature-verified endpoints
- Tenant data strictly isolated via `tenant_id` row-level scoping on all queries
- Call recordings encrypted at rest in Firebase Storage
- No tenant can read another tenant's data under any code path
- Twilio webhook requests validated using X-Twilio-Signature HMAC header

5.x Privacy & Compliance

- Tenant must acknowledge call recording consent before enabling recording feature
- Callers informed of AI agent and recording at call start (configurable greeting)
- Tenant data deletion request fulfilled within 30 days (GDPR-aligned)
- Call transcripts retained for 90 days by default; tenant can configure retention period

5.x Usability

- A business owner with no technical background must be able to configure and launch an agent in under 30 minutes
- Flutter Web dashboard must be fully functional on desktop browsers (Chrome, Firefox, Safari)
- Dashboard must support Arabic RTL layout for Arab business owners

- All error messages in the dashboard must be in the tenant's chosen language

6. Technology Stack

Layer	Technology	Purpose	Cost
Backend framework	Spring Boot 3.x	Core application, REST APIs, WebSocket, scheduling	Free OSS
Language	Java 21	LTS release, virtual threads (Project Loom) for high concurrency WebSocket handling	Free
Database	PostgreSQL 16 + pgvector	All persistent data + RAG vector embeddings for FAQ	DigitalOcean \$15/mo
Cache / Session	Redis 7	Call session state, usage metering, API rate limiting	DigitalOcean \$15/mo
File storage	Firebase Storage	Call recordings, FAQ document uploads	Free tier (5GB)
Telephony	Twilio Programmable Voice + MediaStream	Make/receive calls, real-time audio WebSocket	\$0.013/min + \$1/number/mo
STT (EN/FR/AR)	Deepgram Nova-2	Real-time streaming STT with VAD, low latency	~\$0.0043/min
STT (Swahili)	Intron Sahara v2	First-class Swahili ASR, bilingual SW/EN	Paid API
LLM (primary)	Gemini 2.5 Flash	Best multilingual + African language support, fast	\$0.00015/1K tokens
LLM (fallback)	GPT-4o	Fallback for complex queries	~\$0.005/1K tokens
TTS	ElevenLabs Streaming	Voice cloning, multilingual, low-latency streaming	\$0.0003/char
TTS Arabic	Azure Cognitive Services	High-quality Arabic neural voices (MSA + dialects)	\$0.004/1K chars
TTS Swahili	Google Cloud TTS	Swahili neural voices	~\$0.004/1K chars
AI framework	Spring AI	Unified LLM abstraction, prompt templating, streaming	Free OSS
Frontend	Flutter Web	Business dashboard — single codebase for web/iOS/Android	Free OSS
Payments	Lemon Squeezy	Subscriptions + usage billing; works from Egypt	5% + \$0.50/transaction
Email	SendGrid	Transactional emails (alerts, weekly reports, verification)	Free tier (100/day)
Infrastructure	DigitalOcean Droplet	1× \$12/mo 2vCPU/2GB RAM for Spring Boot	\$12/mo
CI/CD	GitHub Actions + Docker	Build, test, deploy pipeline from push to production	Free (GitHub Actions)

7. Data Models & Database Schema

All tables include standard audit columns: **created_at** **TIMESTAMPTZ**, **updated_at** **TIMESTAMPTZ**. Tenant isolation enforced via **tenant_id** **UUID** foreign key on all tenant-scoped tables. Row-level security (RLS) optional but recommended for production.

Table: tenants

Column	Type / Constraint	Description
id	UUID PRIMARY KEY	Unique tenant identifier
business_name	VARCHAR(255) NOT NULL	Display name of the business
email	VARCHAR(255) UNIQUE NOT NULL	Owner email, used for login
country_code	CHAR(2) NOT NULL	ISO 3166-1 alpha-2 country code
plan	VARCHAR(50) DEFAULT 'trial'	starter growth pro trial
plan_expires_at	TIMESTAMPTZ	NULL for active subscriptions
monthly_minutes_limit	INT DEFAULT 60	Minutes included in plan
minutes_used_this_cycle	INT DEFAULT 0	Reset on billing cycle renewal
lemon_squeezy_customer_id	VARCHAR(100)	External billing reference
status	VARCHAR(20) DEFAULT 'active'	active paused suspended

Table: agents

Column	Type / Constraint	Description
id	UUID PRIMARY KEY	Unique agent identifier
tenant_id	UUID NOT NULL REFERENCES tenants	Owns tenant
name	VARCHAR(100) NOT NULL	Agent display name (e.g. 'Amina')
twilio_phone_number	VARCHAR(20) UNIQUE	E.164 format phone number
default_language	VARCHAR(10) DEFAULT 'fr'	BCP-47 language code
supported_languages	VARCHAR[] DEFAULT '{fr}'	Array of supported language codes
system_prompt	TEXT NOT NULL	LLM system prompt defining agent behavior
greeting_message	TEXT NOT NULL	First message spoken to caller
tts_voice_id	VARCHAR(100)	ElevenLabs or provider-specific voice ID
tts_provider	VARCHAR(50) DEFAULT 'elevenlabs'	elevenlabs azure google
llm_provider	VARCHAR(50) DEFAULT 'gemini'	gemini openai claude
escalation_phrases	TEXT[]	Phrases triggering human transfer
human_transfer_number	VARCHAR(20)	E.164 number for human escalation
operating_hours	JSONB	{ weekday: {start, end}, weekend: null } etc.
is_active	BOOLEAN DEFAULT false	Must be true for calls to be handled
calendar_integration_type	VARCHAR(50)	google calendly webhook none

Column	Type / Constraint	Description
calendar_config	JSONB	Encrypted credentials and config per integration type

Table: calls

Column	Type / Constraint	Description
id	UUID PRIMARY KEY	Unique call identifier
tenant_id	UUID NOT NULL REFERENCES tenants	Owning tenant
agent_id	UUID NOT NULL REFERENCES agents	Agent that handled the call
twilio_call_sid	VARCHAR(100) UNIQUE NOT NULL	Twilio's call identifier
caller_number	VARCHAR(20)	Caller's phone number (may be withheld)
direction	VARCHAR(10) NOT NULL	inbound outbound
language_detected	VARCHAR(10)	BCP-47 code of detected language
started_at	TIMESTAMPTZ NOT NULL	Call start timestamp
ended_at	TIMESTAMPTZ	Call end timestamp; NULL while active
duration_seconds	INT	Total call duration
outcome	VARCHAR(50)	booking_made faq_answered transferred voicemail abandoned
transcript	TEXT	Full conversation transcript
recording_url	VARCHAR(500)	Firebase Storage URL if recording enabled
sentiment	VARCHAR(20)	positive neutral negative (post-call LLM analysis)
booking_id	UUID REFERENCES bookings	Linked booking if outcome = booking_made

Table: bookings

Column	Type / Constraint	Description
id	UUID PRIMARY KEY	Unique booking identifier
tenant_id	UUID NOT NULL REFERENCES tenants	Owning tenant
agent_id	UUID NOT NULL REFERENCES agents	Agent that created the booking
call_id	UUID REFERENCES calls	Source call
caller_name	VARCHAR(255)	Name provided by caller
caller_phone	VARCHAR(20)	Caller phone number
service_type	VARCHAR(255)	Type of service booked
booked_at	TIMESTAMPTZ NOT NULL	When the appointment was made
appointment_at	TIMESTAMPTZ NOT NULL	When the appointment is scheduled
external_event_id	VARCHAR(255)	Google Calendar / Calendly event ID
status	VARCHAR(20) DEFAULT 'confirmed'	confirmed cancelled rescheduled
confirmation_sent	BOOLEAN DEFAULT false	SMS confirmation sent flag

Table: agent_documents

Column	Type / Constraint	Description
id	UUID PRIMARY KEY	Document identifier
agent_id	UUID NOT NULL REFERENCES agent	Ownning agent
filename	VARCHAR(255) NOT NULL	Original uploaded filename
storage_url	VARCHAR(500) NOT NULL	Firebase Storage URL
status	VARCHAR(20) DEFAULT 'processing'	processing indexed failed
chunk_count	INT	Number of vector chunks created

Table: document_chunks

Column	Type / Constraint	Description
id	UUID PRIMARY KEY	Chunk identifier
document_id	UUID NOT NULL REFERENCES document	Parent document
agent_id	UUID NOT NULL REFERENCES agent	For direct lookup without join
content	TEXT NOT NULL	Text content of this chunk
embedding	vector(1536)	pgvector embedding (OpenAI text-embedding-3-small)
chunk_index	INT NOT NULL	Sequence number within document

8. API Specification

All endpoints use **HTTPS**. Authentication via **Bearer JWT** in Authorization header unless marked [Public].
Twilio webhook endpoints verified via **X-Twilio-Signature** HMAC-SHA1.

Authentication — /api/v1/auth

Method	Path	Description	Auth
POST	/register	Register new tenant account	Public
POST	/login	Authenticate, receive JWT + refresh token	Public
POST	/refresh	Rotate access token using refresh token	Public
POST	/logout	Invalidate refresh token	Authenticated
POST	/forgot-password	Send password reset email	Public
POST	/reset-password	Complete password reset with token	Public

Agents — /api/v1/agents

Method	Path	Description	Auth
GET	/	List all agents for authenticated tenant	Authenticated
POST	/	Create new agent (provisions Twilio number)	Authenticated
GET	/ {id}	Get agent configuration	Authenticated
PUT	/ {id}	Update agent configuration	Authenticated
DELETE	/ {id}	Deactivate and release Twilio number	Authenticated
POST	/ {id}/activate	Activate agent (enables call handling)	Authenticated
POST	/ {id}/test	Initiate test call to tenant's verified number	Authenticated
POST	/ {id}/documents	Upload FAQ document for RAG indexing	Authenticated

Calls — /api/v1/calls

Method	Path	Description	Auth
GET	/	List calls with filters (date, agent, outcome, language)	Authenticated
GET	/ {id}	Get full call details including transcript	Authenticated
POST	/outbound	Initiate outbound call from an agent	Authenticated

Twilio Webhooks — /webhooks/twilio

Method	Path	Description	Auth
POST	/voice	Inbound call webhook (Twilio → platform)	[Twilio Signed]
POST	/voice/stream	Media stream events (connected, stop)	[Twilio Signed]
POST	/status	Call status callback (completed, failed)	[Twilio Signed]

Analytics — /api/v1/analytics

Method	Path	Description	Auth
GET	/summary	Tenant call summary for date range	Authenticated
GET	/by-language	Call breakdown by language detected	Authenticated
GET	/by-outcome	Call breakdown by outcome classification	Authenticated
GET	/peak-hours	Heatmap data: calls by hour × day of week	Authenticated

Billing — /api/v1/billing

Method	Path	Description	Auth
GET	/usage	Current minute usage this billing cycle	Authenticated
GET	/invoices	List historical invoices	Authenticated
POST	/webhooks/ls	Lemon Squeezy subscription webhook	[LS Signed]

Bookings — /api/v1/bookings

Method	Path	Description	Auth
GET	/	List bookings with filters	Authenticated
GET	/ {id}	Get booking detail	Authenticated
DELETE	/ {id}	Cancel booking (notifies caller via SMS)	Authenticated

9. Security Requirements

Authentication & Authorization

- JWT access tokens with 15-minute expiry; refresh tokens with 30-day expiry stored in HttpOnly cookies
- Refresh token rotation on every use; old token invalidated immediately
- Spring Security method-level authorization (@PreAuthorize) on all service methods
- All tenant-scoped repository queries include WHERE tenant_id = :tenantId — enforced at service layer
- Twilio webhook signature validation using HMAC-SHA1 of URL + POST body with Twilio auth token
- Lemon Squeezy webhook signature validation via X-Signature header

Data Protection

- PostgreSQL connection over SSL/TLS mandatory in production
- calendar_config JSONB field encrypted at application layer before storage (AES-256)
- Call recordings stored in Firebase Storage with tenant-scoped access rules
- Passwords hashed with BCrypt, cost factor 12
- Sensitive environment variables managed via DigitalOcean App Platform secrets or SOPS

API Security

- Rate limiting on all public endpoints: 10 requests/minute per IP on auth endpoints
- CORS restricted to Flutter Web dashboard domain and Twilio webhook IPs
- HTTPS enforced via Let's Encrypt certificate with HSTS header
- SQL injection prevented via Spring Data JPA parameterized queries only — no native query string concatenation
- Request body size limits enforced (max 10MB on document upload, 1KB on standard API)

Operational Security

- Spring Boot Actuator endpoints (/actuator/**) protected behind separate management port (8081) with IP allowlist
- Database credentials rotated quarterly; access logged via DigitalOcean audit logs
- Failed login attempts tracked in Redis; account locked after 10 consecutive failures for 15 minutes
- All production logs must not contain PII (caller phone numbers masked as +XX***XXXX in logs)

10. Infrastructure & Deployment

Total fixed infrastructure cost at MVP: \$42/month (Droplet \$12 + PostgreSQL \$15 + Redis \$15). Variable AI API costs scale with usage and are covered by tenant billing.

10.1 Infrastructure Components

Component	Provider	Spec	Monthly Cost
Spring Boot App Server	DigitalOcean Droplet	2 vCPU / 2GB RAM / 50GB SSD	\$12
PostgreSQL Database	DigitalOcean Managed DB	1 vCPU / 1GB RAM, daily backups, SSL	\$15
Redis Cache	DigitalOcean Managed Redis	1GB memory, persistence disabled	\$15
File Storage	Firebase Storage	Free tier: 5GB storage, 1GB/day download	\$0
Domain & SSL	Namecheap + Let's Encrypt	Annual domain + free cert via Certbot	~\$1
Email	SendGrid	Free tier: 100 emails/day	\$0
Total (fixed)			\$43/month

10.2 CI/CD Pipeline

Push to main	GitHub Actions triggers build workflow
Build	<code>./gradlew build</code> — compiles, runs unit tests, generates JAR
Docker build	<code>docker build -t sauti-backend:\$SHA .</code> — creates image
Push	<code>docker push</code> to Docker Hub or GitHub Container Registry
Deploy	SSH to DigitalOcean Droplet → <code>docker pull</code> → <code>docker-compose up -d</code>
Health check	GitHub Actions waits for <code>/actuator/health</code> to return 200 before marking success

11. Build Sequence & Milestones

As a solo developer, the build sequence is optimized for the fastest path to a paying customer. Each phase ends with something shippable.

Phase 1 — Foundation	Weeks 1–3
• Spring Boot project setup with all modules scaffolded • module-auth: JWT login, registration, refresh token rotation • module-tenant: account creation, plan model • PostgreSQL schema: tenants, agents, calls tables • Basic Flutter Web: login screen + dashboard skeleton • Milestone: A business owner can register and log in	
Phase 2 — Agent Config & Twilio	Weeks 4–6
• module-agent: full CRUD for agent configuration • Twilio phone number provisioning on agent creation • Inbound Twilio webhook → TwiML response (static 'Hello' for now) • Flutter Web: agent configuration form (name, language, greeting, system prompt) • Milestone: Creating an agent provisions a real phone number; calling it plays a static greeting	
Phase 3 — Live Call Pipeline	Weeks 7–10
• Twilio Media Streams WebSocket connection in module-call • Deepgram real-time STT integration with VAD • Spring AI + Gemini 2.5 Flash: first LLM response from system prompt • ElevenLabs streaming TTS → audio back to Twilio • Redis conversation history per call session • Language detection middleware • Milestone: A caller can have a full AI conversation with the agent in French or English	
Phase 4 — Booking & Calendar	Weeks 11–13
• module-calendar: Google Calendar OAuth integration • Agent can check availability and create bookings during a call • Twilio SMS confirmation after booking • bookings table + booking outcome in calls table	

- Flutter Web: bookings list view
- Milestone: A clinic can receive calls and book appointments automatically

Phase 5 — Billing & Launch

Weeks 14–16

- module-billing: Lemon Squeezy webhook + plan enforcement
- Usage metering: per-minute tracking in Redis, synced to PostgreSQL
- Email alerts at 80% / 100% usage via SendGrid
- Flutter Web: billing page, invoice history, plan upgrade flow
- Intron Sahara v2 integration for Swahili
- Arabic TTS via Azure Cognitive Services
- Milestone: Platform is billable; first paying customer can be onboarded

Phase 6 — Analytics & Polish

Weeks 17–20

- module-analytics: call log aggregation, outcome classification
- Flutter Web: analytics dashboard with charts
- Weekly summary email (Spring Scheduler + SendGrid)
- Call recording + transcript search
- Agent test call simulator in dashboard
- RAG pipeline: FAQ document upload + pgvector indexing
- Milestone: Platform is self-service; business owners need no support to get started

12. Risks & Mitigations

Level	Risk	Mitigation
HIGH	Twilio WebSocket latency spikes	Use Java 21 virtual threads (Project Loom) for non-blocking WebSocket handling; n
HIGH	Swahili STT accuracy below expectations	Test Intron Sahara v2 early with real Swahili speakers; maintain Whisper large-v3 a
HIGH	LLM hallucination in agent responses	Ground all agent responses in tenant system prompt and RAG-retrieved FAQ chun
MEDIUM	Twilio API cost overruns	Enforce hard per-tenant minute caps in Redis; auto-pause agent when limit reached
MEDIUM	Single droplet becomes bottleneck	Monitor concurrent WebSocket connections; vertical upgrade path clear (\$12→\$24
MEDIUM	Arabic dialect recognition failures	Start with Egyptian Arabic (MSA-adjacent, largest user base); expand to other diale
LOW	Payment processor regional restrictions	Lemon Squeezy and Paddle both verified to work from Egypt; test payment flow be
LOW	GDPR / local data regulation compliance	Call transcripts stored in EU-region Firebase by default; consent captured in greetin

13. Glossary

Term	Definition
BCP-47	A standard for language tags (e.g. 'fr' for French, 'ar-EG' for Egyptian Arabic).
CRDT	Conflict-free Replicated Data Type — a data structure for distributed conflict resolution.
Deepgram Nova-2	A real-time STT API with strong multilingual support and low latency.
E.164	International phone number format: +[country code][number], e.g. +2201234567.
ElevenLabs	A TTS API offering voice cloning and streaming audio generation.
Gemini 2.5 Flash	Google's fast, cost-efficient LLM with strong African language support.
Intron Sahara v2	The world's first bilingual Swahili-English ASR model, available as an API.
LemonSqueezy	A merchant of record payment platform supporting global sellers including Egypt.
Loom (Project)	Java 21 feature providing virtual threads for high-concurrency I/O without thread pools.
MJML	A framework for writing responsive HTML email templates.
Modular Monolith	A single deployable application internally structured into bounded module packages.
pgvector	A PostgreSQL extension for storing and querying vector embeddings.
RAG	Retrieval-Augmented Generation: injecting relevant document chunks into LLM context.
SM-2	A spaced repetition algorithm used in flashcard applications (not relevant to Sauti but mentioned in Nu
Spring AI	A Spring framework abstraction for LLM integration with prompt templates and streaming.
Twiml	Twilio Markup Language: XML instructions telling Twilio how to handle a call.
Twilio Media Streams	A Twilio feature that streams raw audio over WebSocket for real-time processing.
VAD	Voice Activity Detection: identifying when a speaker has finished speaking.
Webhook	An HTTP callback sent by a third-party service to notify your application of an event.